

Case 23: Spatially varying fuel quadrants

1 Verification purpose

Case ID: CASE23_FQD. This case verifies raster lookup and fire-behavior coupling when a propagating front crosses four fuel-model regions. Flat terrain and zero wind isolate fuel identity. The verification question is whether every quadrant's maximum spread rate is within 10 percent of its fuel-specific reference.

The scientific setup and historical expected response are reconstructed from Section 3.2 of the supplied *ELMFIRE Guide*. The Guide is used as reference provenance, not as an instruction or a substitute for evidence. Configuration statements below are checked against the generated GeoTIFFs and namelist; current results and decisions come only from case-local outputs and generated metrics. Where the generated experiment differs from the historical example, the report states that difference explicitly.

2 Mathematical formulation and numerical implementation

The level-set normal speed is evaluated from the fuel model at each active cell. Because the environmental forcing is uniform, changes in V_n across the domain should follow only changes in the fbfm raster. Correct row orientation and quadrant masking are therefore part of the test.

At each active cell the fuel code selects a complete Rothermel parameter row, and the local no-wind/no-slope rate is

$$R_0(x, y) = \frac{I_R(x, y) \xi(x, y)}{\rho_b(x, y) \epsilon(x, y) Q_{ig}(x, y)}. \quad (1)$$

Thus the raster lookup changes the full nonlinear fuel response, not a post-hoc scalar multiplier. The formulation follows Andrews (2018); the Guide supplies the four BEHAVE reference values.

3 Derivation of expected behavior

The independent regional targets are 0.080, 0.472, 1.492, and 0.871 m min⁻¹ for Fuel Models 8, 7, 4, and 2 in the northwest, northeast, southwest, and southeast quadrants, respectively. These values also predict a visibly noncircular front after it enters all four regions. The Guide historically reported ELMFIRE rates of 0.080, 0.551, 1.974, and 0.872 m/min for FB8, FB7, FB4, and FB2, noting discrepancies for FB7 and FB4. Those historical outputs are preserved here as context only; the generated metric table below is the current evidence and is never replaced by those values.

All four references use the no-wind/no-slope Rothermel equation $R_0 = I_R \xi / (\rho_b \epsilon Q_{ig})$ with $Q_{ig} = 250 + 1116 M_f$ and the generated moisture vector (0.06, 0.07, 0.08, 0.60, 0.90). Substitution of each generated standard fuel model gives

fuel model	I_R (kW m ⁻²)	R_0 (m min ⁻¹)
8	175.566	0.079552
7	406.479	0.470579
4	2272.300	1.493256
2	669.483	0.871756

and hence the report references 0.080, 0.472, 1.492, and 0.871 m min^{-1} , respectively. For each fuel model, $e_R = |R_{\max} - R_{\text{ref}}|/R_{\text{ref}}$; the table substitutes the corresponding row above and the prescribed relative-error limit.

4 Verification metrics and acceptance criteria

The maximum finite vs in each quadrant is compared with its corresponding target. Regional maxima preserve the local head-fire response and avoid mixing physically distinct fuels. Every component uses 10 percent relative tolerance, and PASS requires all four components.

5 Simulation configuration and predicted outcome

Four equal quadrants on a 126×126 , 30 m grid contain Fuel Models 8, 7, 4, and 2. Terrain is flat, wind is zero, moisture is uniform, and ignition is central. The fastest extension is predicted in Fuel Model 4 and the slowest in Fuel Model 8.

Parameter	Configured value	Units	Role in the experiment
Grid	126×126 at 30	m per cell	Resolves the whole spatial response on the generated north-up raster
Projection	EPSG:32635	—	Fixes the map-coordinate and row-orientation convention
Time control	$\Delta t = 1$, $t_{\text{stop}} = 259200$	s	Whole-second step; stop time is an exact timestep multiple
Meteorology interval	3600	s	Defines weather-band timing and stalled-final compatibility
Live moisture	herb 60; woody 90	percent	Holds live-fuel damping fixed while the named mechanism is isolated
Evidence	prepared inputs; final case outputs	—	Separates configuration provenance from acceptance evidence

Figure 1 is generated mechanically from the prepared case-local GeoTIFF inputs. It identifies the selected source artifact and representative variant; the left panel shows a fuel, terrain, moisture, wind, canopy, or other controlling field selected for this experiment, and the right panel shows the cells selected by the non-positive initial level-set field. No values in this figure are redrawn or inferred from the expected response.

Whole-domain input configuration — data; WS=0 mph, WD=0 deg

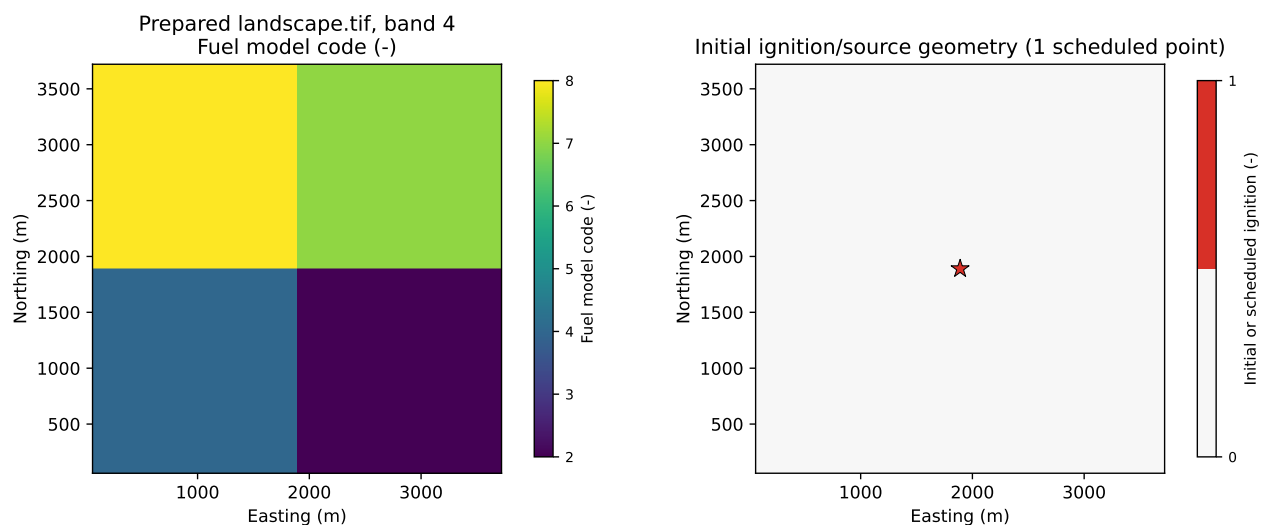


Figure 1: Prepared input conditions from case-local generated rasters. Coordinates are in metres; the colorbar identifies the displayed field and its physical units or category. The ignition panel is derived directly from $\phi \leq 0$.

6 Actual simulation results

The required evidence is generated only from case-local ELMFIRE outputs. The current postprocessing status is **PASS**; 4 of 4 required metric rows pass across 1 variant(s). A missing output is reported as NOT RUN or NOT EVALUABLE and is not interpreted as verification evidence.

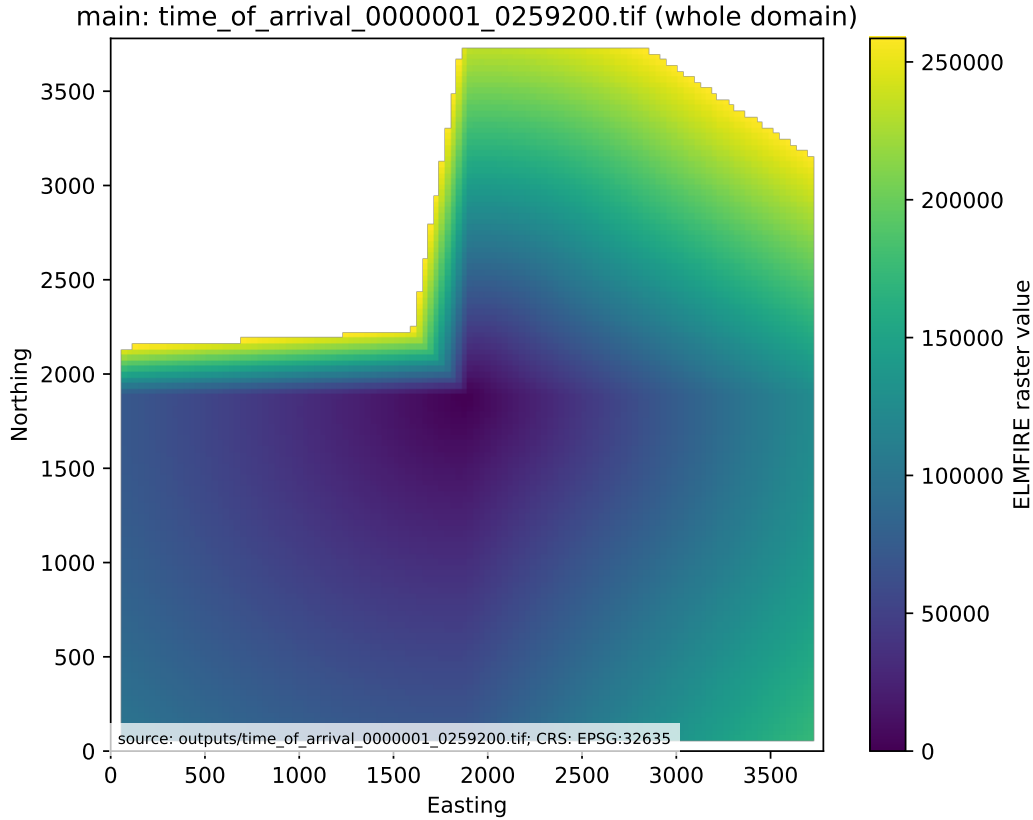


Figure 2: Final whole-domain ELMFIRE time of arrival for the displayed variant, in seconds. The figure annotation records the selected case-local GeoTIFF path and coordinate reference system; the generated metrics table, not visual appearance, determines acceptance.

7 Calculated metrics and verification decision

Variant / metric	Expected	Calculated	Status
main / FB8 head ROS	0.08 m/min	0.0795524 m/min	PASS
main / FB7 head ROS	0.472 m/min	0.470579 m/min	PASS
main / FB4 head ROS	1.492 m/min	1.49326 m/min	PASS
main / FB2 head ROS	0.871 m/min	0.871756 m/min	PASS

The displayed overall decision is **PASS**. When `outputs/metrics.json` exists, `scripts/postprocess.py` generates every table value and status; when it is absent, the standalone wrapper renders NOT EVALUABLE with no invented rows.

8 Technical interpretation

A quadrant swap indicates raster orientation or mask errors; a fuel-specific discrepancy indicates fuel-table selection or fire-behavior coupling. No scientific conclusion should be drawn when the status is NOT RUN or NOT EVALUABLE.

9 Scope and limitations

The test samples four fuel models under one moisture state and does not validate the empirical fuel parameters. This case provides implementation verification evidence only.